

Mellow LLL Buffer For RRF



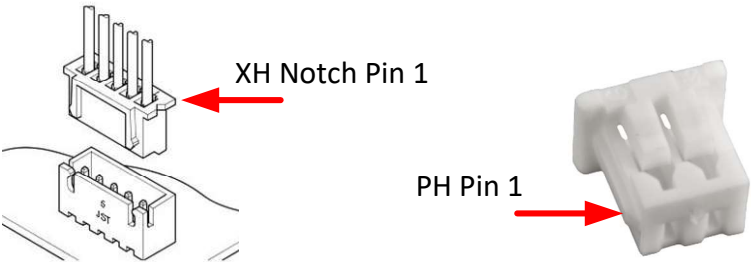
- Items required for the buffer
- 1 x Mellow LLL Buffer
  - 2 x JST PHR-2
  - 1 x JST XH 3-Way
  - 1 x JST XH 4-Way

Note you will need cable and connector for your controll board

Control Board connections required

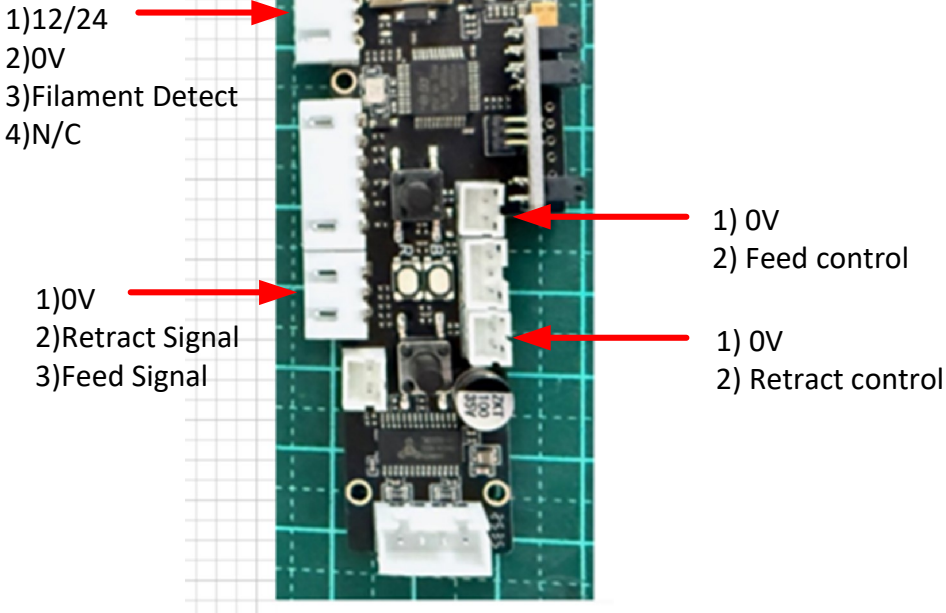
3 inputs,, Filament Detect , Button control Load/unload

2 outputs, control buffer load/unload



Example of connections to a Duet 6HC

| From Buffer     | Operation              |         | TO Duet Pin |
|-----------------|------------------------|---------|-------------|
| Retract Control | Retract Filament       | control | IO_7_out    |
| Feed Control    | Feed Filament          | control | IO_8_out    |
| Filament Detect | Check Filament Loaded  | signal  | IO_8_in     |
| Feed Signal     | Button signal Feed     | signal  | IO_6_in     |
| Retract Signal  | Button singnal Retract | signal  | IO_7_in     |



# Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

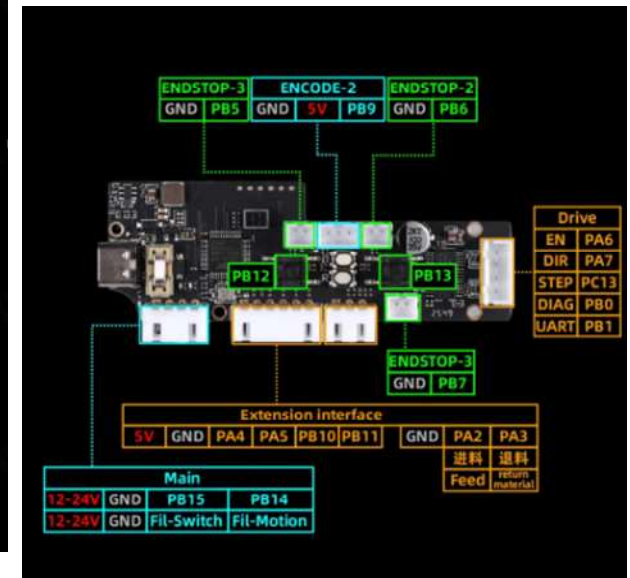
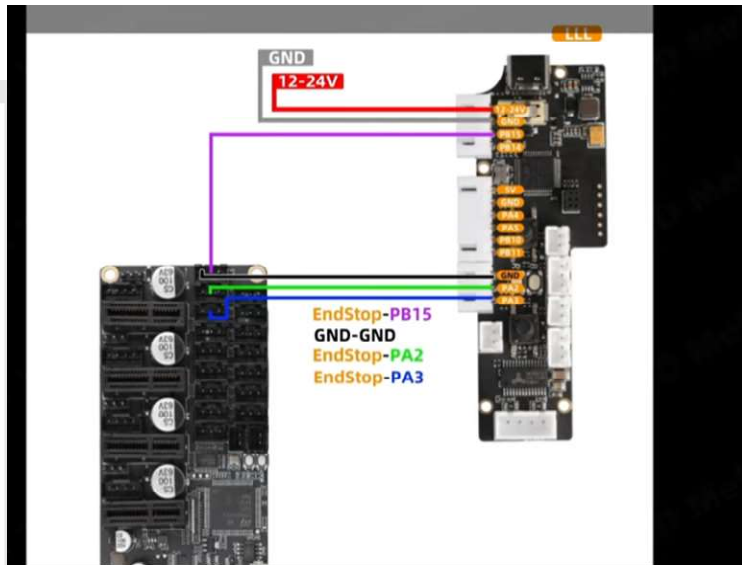
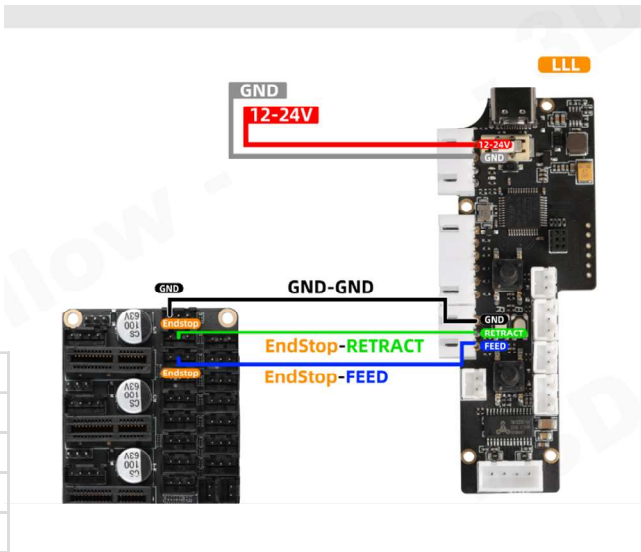
| Buffer Pin      | Trigger Signal | Action Performed                      | Duration                              |
|-----------------|----------------|---------------------------------------|---------------------------------------|
| Feed Control    | Low Level      | Buffer continuously feeds filament    | Executes while the signal is held low |
| Retract Control | Low Level      | Buffer continuously retracts filament | Executes while the signal is held low |
|                 |                |                                       |                                       |

| Button         | Operation    | Signal Output (Buffer Pin)            | Signal Type | Duration                                            |
|----------------|--------------|---------------------------------------|-------------|-----------------------------------------------------|
| Feed Button    | Click        | FEED pin outputs a high-level pulse   | High Level  | Automatically returns to low level after 3 seconds  |
| Feed Button    | Press & Hold | Continuous feeding                    | High Level  | Until the button is released                        |
| Retract Button | Click        | RETRACT pin outputs a low-level pulse | Low Level   | Automatically returns to high level after 3 seconds |
| Retract Button | Press & Hold | Continuous retraction                 | Low Level   | Until the button is released                        |

Note: The buffer stops the action when the signal returns to a high level.

## How to up date Firmware

<https://mellow.klipper.cn/en/docs/ProductDoc/ExtensionBoard/fly-buffer-plus/update>



## Signal Output Description #

| Button                   | Operation  | Signal Output (Buffer Pin)            | Signal Type | Duration                                            |
|--------------------------|------------|---------------------------------------|-------------|-----------------------------------------------------|
| Feed Button (FEED)       | Click      | FEED pin outputs a high-level pulse   | High Level  | Automatically returns to low level after 3 seconds  |
| Feed Button (FEED)       | Long Press | Continuous feeding                    | High Level  | Until the button is released                        |
| Retract Button (RETRACT) | Click      | RETRACT pin outputs a low-level pulse | Low Level   | Automatically returns to high level after 3 seconds |
| Retract Button (RETRACT) | Long Press | Continuous retraction                 | Low Level   | Until the button is released                        |

## Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

| Buffer Pin | Trigger Signal | Action Performed                      | Duration                              |
|------------|----------------|---------------------------------------|---------------------------------------|
| PB5        | Low Level      | Buffer continuously feeds filament    | Executes while the signal is held low |
| PB6        | Low Level      | Buffer continuously retracts filament | Executes while the signal is held low |

Note: The buffer stops the action when the signal returns to a high level.